BIOLOGY 5090

Paper 2

MARK SCHEME

Maximum Mark: 80

Question	Answer	Marks	Guidance
1(a)	warm ; water ; oxygen ;	**3**	
1(b)	1 water absorbed / goes in ; 2 seed increases + in size / volume ; 3 enzymes become active AW; 4 digestion / hydrolysis / breakdown of starch / protein / lipids ; 5 respiration ; 6 testa / seed coat / seed wall softens / splits AW ; 7 radicle / root + grows (out) ; **max 5**	**5**	
1(c)(i)	embryo ;	**1**	
1(c)(ii)	X / plumule /shoot + negative gravitropism / grows against gravity ; Y / radicle / root + positive gravitropism / grows towards gravity ;	**2**	

Question	Answer	Marks	Guidance
2(a)(i)	minerals ; vitamins ; fibre ; water ; **max 3**	**3**	
2(a)(ii)	energy / energy store ; electrical insulation in neurones ; (thermal) insulation ; mechanical protection; components of cell membranes ; **max 2**	**2**	
2(b)(i)	60 / 59.8 / 59.81 ;;	**2**	
2(b)(ii)	15 / 15.3 / 15.33 ;;	**2**	
2(c)	less deforestation / more land/habitat available for wildlife ; more biodiversity / less monoculture ; less extinction ; reduced pollution + from insecticides / herbicides / fertilisers ; less erosion / loss of soil ; less flooding ; less methane production + reduced AW global warming / climate change ; **max 3**	**3**	

Question	Answer	Marks	Guidance
3(a)(i)	A: ribosome ;	**1**	
	B: plasmid ;	**1**	
3(a)(ii)	cytoplasm ; cell membrane ; DNA / RNA / genetic material / chromosome ; **max 2**	**2**	
3(a)(iii)	antigens / proteins ; on surface AW of bacteria ORA ; recognised as foreign ORA ; antibody specific to the antigen ; **max 3**	**3**	
3(b)(i)	antibiotic, moves / diffuses into the agar / bacteria ;	**1**	
3(b)(ii)	antibiotic 1: only effective against strain L;	**1**	
	antibiotic 5: is effective against all four strains ;	**1**	
3(c)(i)	N	**1**	
3(c)(ii)	1 variation in population ; 2 N has greatest resistance ; 3 best adapted (to environment) ; 4 they (are more likely to) survive ; 5 they (are more likely to) reproduce ; 6 beneficial alleles passed to next generation / offspring ; 7 natural selection ; **max 5**	**5**	

Q4.

Answer	Marks	Guidance
asexual ;	1	
advantages rapid AW ; many new plants formed / greater yield ; only one parent plant needed ; plants have same desirable AW + feature / allele / named useful feature ; ripen at the same time so easier to harvest ; **max 2** *disadvantages* have same undesirable AW + feature / allele / named useful feature ; not adapted if change in environment AW ; ripen at the same time so price goes down / not a constant supply to market ; **max 2**	4	
with hummingbirds : sexual reproduction / pollination / fertilisation ; seeds formed ; humans / consumers + don't like / find inedible AW ; reduced income / business / economy AW ; hummingbird introduces diseases ; hummingbird + competes with other species ; (competes) for named resource ; **max 4**	4	
protein ; ⟶ amino acids ;	2	

Question	Answer	Marks	Guidance
5(a)(i)	fungi / fungus **+** *Rhizomucor*	**1**	
5(a)(ii)	cell wall made of chitin ; heterotrophic / saprotrophic / saprophytic ; filamentous / hyphae / mycelium ; eukaryotic; **max 2**	**2**	

Question	Answer	Marks	Guidance
5(b)(i)	(pH) 2 / 3 + denatured ; active site of enzyme deformed / changed shape ; substrate no longer fits the active site ; optimum (pH) at 4.5 ; substrate and active site complementary AW ; ES complexes form ; starts to denature above 4.5 ; **max 4**	**4**	
5(b)(ii)	line with optimum activity at pH 4.5 + drawn above the original ;	**1**	
5(b)(iii)	fruit juice production ;	**1**	

Question	Answer	Marks	Guidance
6(a)(i)	(P) chromosome	**1**	
6(a)(ii)	(Q) nucleotide	**1**	
6(a)(iii)	C	**1**	
6(b)	(DNA) codes for proteins / polypeptides ; order of bases / base sequence ; determines sequence of amino acids; example protein;	**4**	

Q7.

Answer	Marks	Guidance
(**P**) Bowman's capsule ; (**Q**) loop of Henle ; (**R**) collecting duct ;	3	
kidney ;	1	
(urinary) bladder ;	1	
toxic / toxin ;	1	
glucose **1** enters Bowman's capsule / enters nephron / removed from blood AW ; **2** by filtration / filtered; **3** back into blood / reabsorbed + into blood ; **4** AVP ; *protein* **5** doesn't enter Bowman's capsule / doesn't enter nephron / no filtration / not removed from blood AW ; **6** protein + large AW ; **max 5**	5	
ion concentration + decrease ; more water / higher volume of water / excess water + in fluid **S** / urine / excreted; homeostasis / to keep concentration of water in blood constant AW;	3	